UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Chapter 1: Information Technology & Computer and System

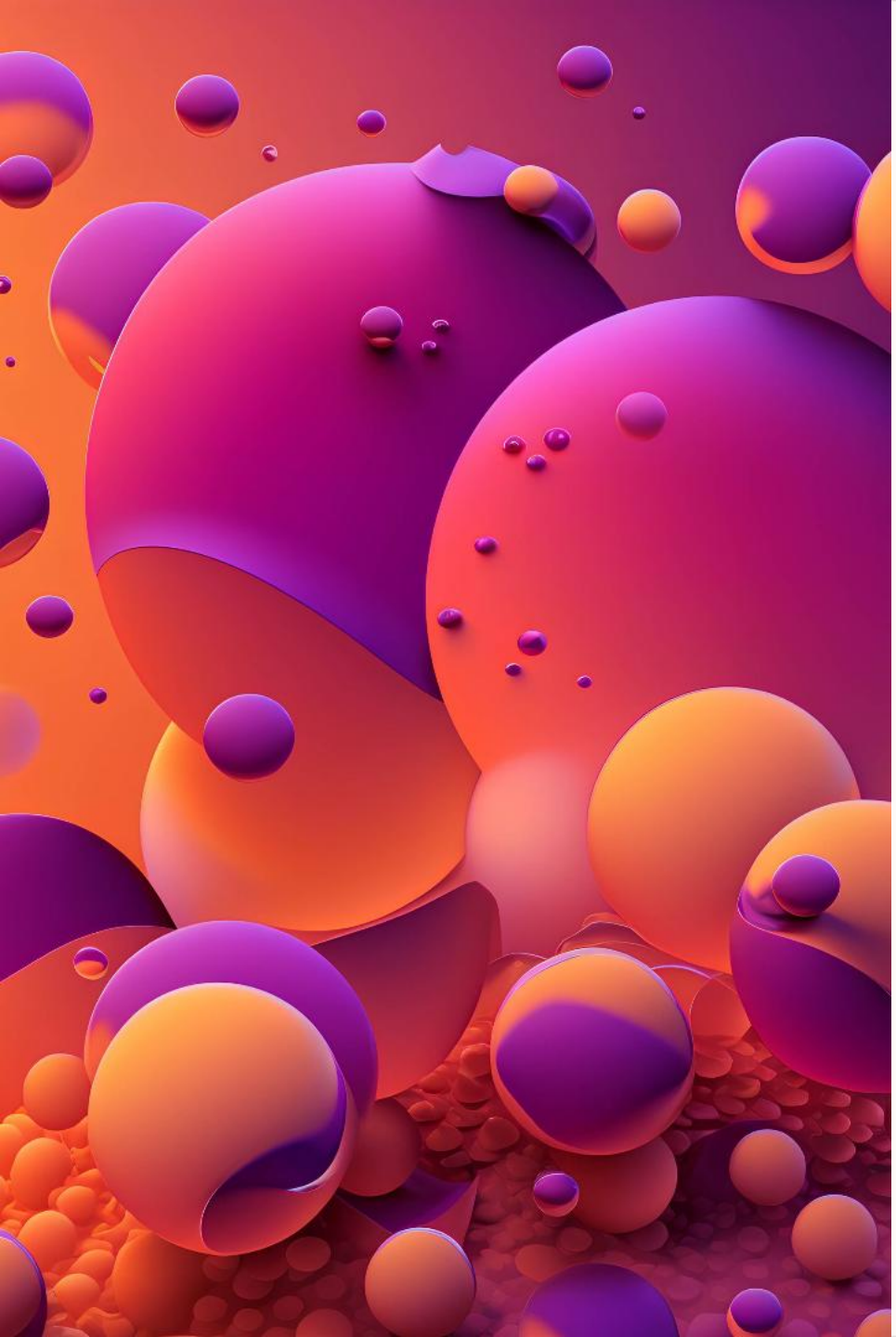


Innovating Solutions

UTM JOHOR BAHRU

Objectives

- Explain the meaning of Information Technology (IT)?
- Explain the computer?
- Explain the system?



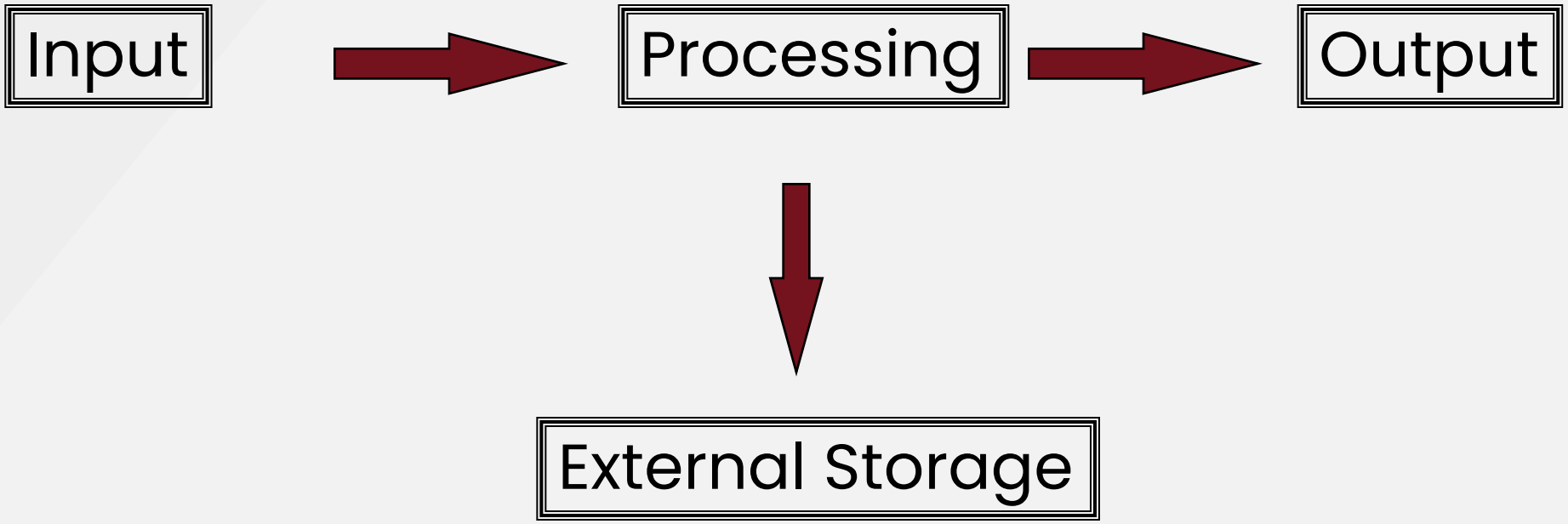
ICT Overview

Information and Communication Technology (ICT) is the backbone of modern-modern-day technological advancements. It covers a wide range of applications, applications, from telecommunication to computer networking. ICT relies on a relies on a combination of **hardware, software, and data** to deliver efficient efficient **communication**.

What is computer?

- COMPUTER :- A computer is a programmable electronic device. It takes data through input devices. It processes the data according to a sequence of instructions provided in the form of a program. The result is called the output and given through some output device.

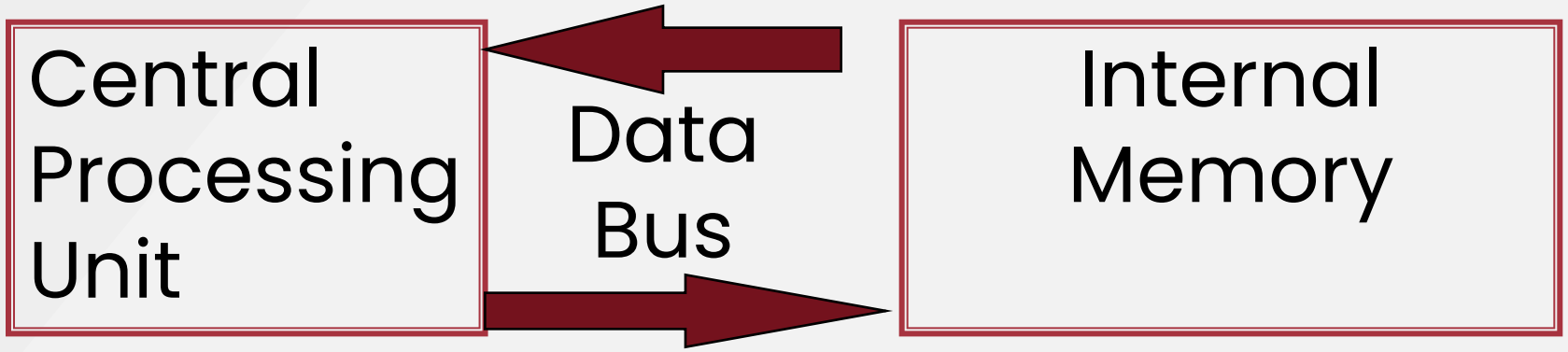
Computer Operations



Hardware

Computer hardware is the collection of physical elements that constitutes a computer system. Computer hardware refers to the physical parts or components of a computer such as the monitor, mouse, keyboard, computer data storage, hard drive disk (HDD), system unit (graphic cards, sound cards, memory, motherboard and chips), etc. all of which are physical objects that can be touched.

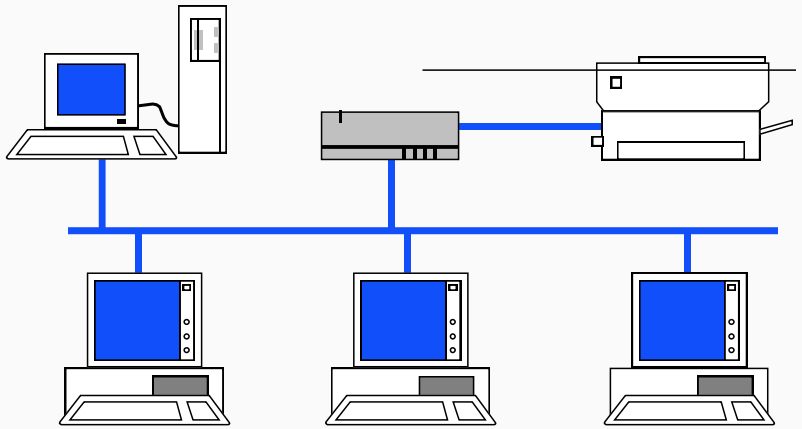
Basic Processing Cycle



How Information Is Stored

- Memory consist of switches which can be either on or off - Off=0 On=1
- Each on/off switch is called a **bit**
- Eight bits make up a **byte**
- It takes one byte to store a character
  - Character can be letter, space, punctuation, etc.
  - ASCII code used

Networks



- Connects computers
- LAN - Local Area
- WAN - Wide Area
- Wireless
- Allows sharing of programs, files, printers, etc.
- Server is “main” computer

Programs

- Set of instructions to the computer
- Programming languages
  - Machine language
  - Assembly language
  - Procedural languages
    - Basic, Fortran, Cobol
  - Object oriented languages
    - Visual Basic, C++, C#, Java

Systems Software

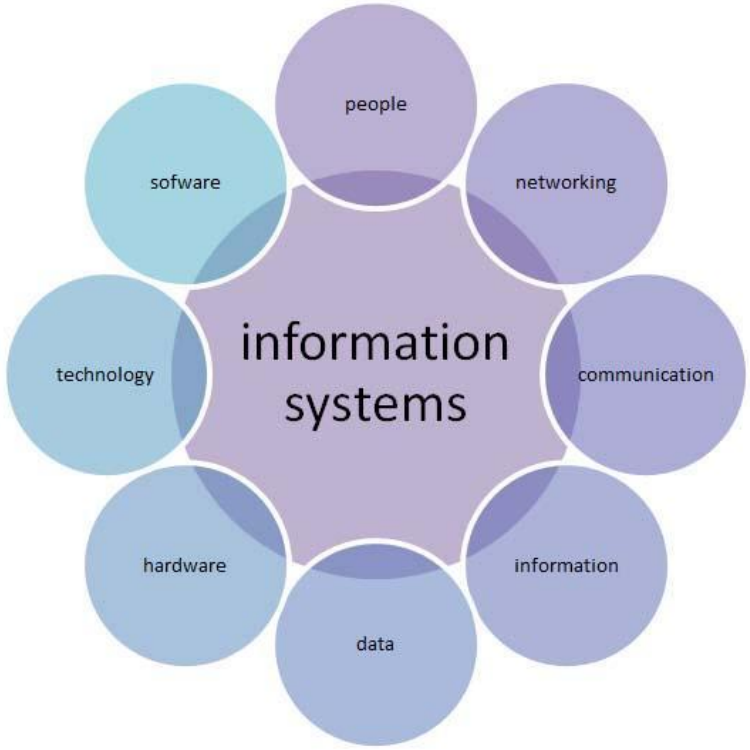
- Run fundamental operations
  - Loading and running programs
  - Saving and retrieving data
  - Communicating with printers, modems, etc.
- Examples of systems software
  - DOS
  - Windows 3.1, 95, 98, Me, 2000, and XP
  - Unix
  - Linux

WHAT IS INFORMATION SYSTEMS (IS)?

A system which assembles, stores, processes and delivers information relevant to an organization, in such a way that the information is accessible and useful to those who wish to use it, including managers, staff, clients and citizens.

An IS is human activity (social system) which may or may not involve the use of computer systems

…Buckingham et al. (1987)



History of ICT Development



Ages of Basic Information Technology

Pre-Mechanical	Mechanical	Electromechanical	Electronic
between 3000B.C. and 1450A.D -alphabet -petroglyph -abacus	between 1450 and 1840 lots of different machines created but have not yet gotten to a machine that can do more than one type of calculation in one -Pascaline -Slide rule -Difference Engine	between 1840 and 1940 beginnings of telecommunication -telegraph -telephone -Mark I	between 1940 and right now Generations: 1st – vacuum tube, punch cards 2nd – transistors, magnetic tape, FORTRAN, COBOL 3rd – integrated circuit, metal oxide 4th – CPU, GUI

ELECTRONIC

First Generation (Vacuum Tubes)

information is sent by electric current (electron movement) whose speed is close to light velocity (300,000 km/second)

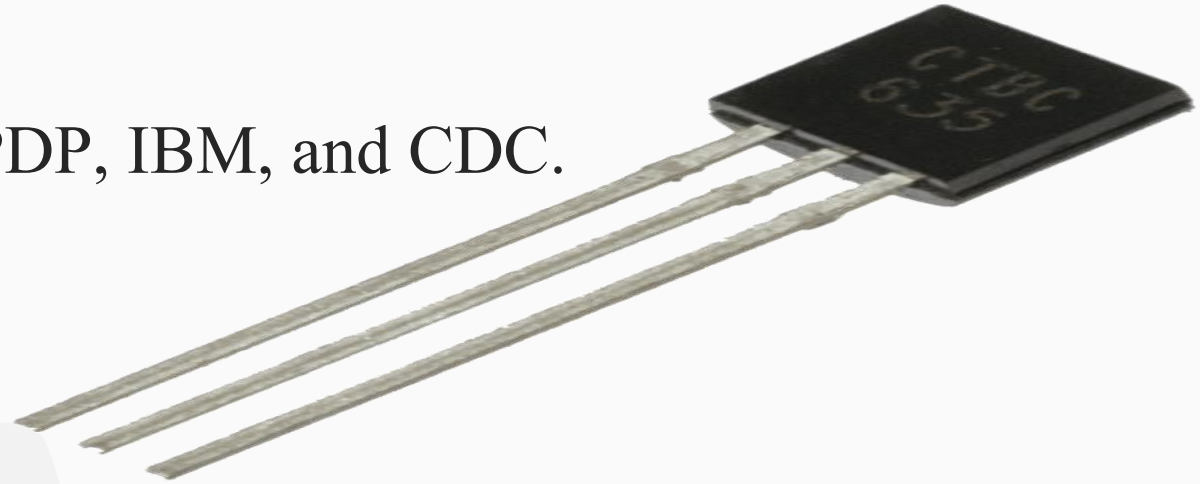
Was programmed with 6000 multiposition electric switches

programs and data were stored in separate memories, so that data entry and alteration were not a boring job



Second Generation Computer (Transistor): 1955-1965

- The transistor era was marked by the innovation on hardware and some program design methodology (software technology).
- It used transistor as vacum tubes substitute.
- “High level” programming language did not depend on independent machine such as ALGOL, COBOL, FORTRAN.
- Some of the computers were PDP, IBM, and CDC.



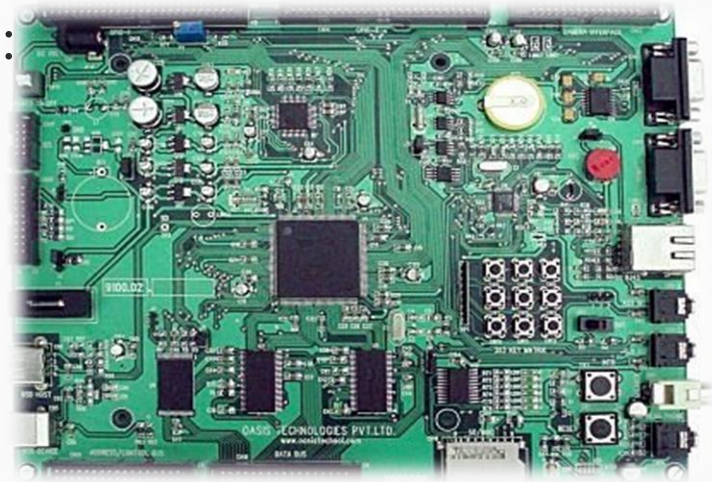
Third Generation Computer (Integrated Circuit): 1965-1980

In this era some of the development were:

- IC was used as transistor substitute, so the size was smaller.
- Semiconductor memory was used as main memory.
- *Microprogramming* technique, makes CPU design simpler and more flexible.
- Parallel processing was used, so counting was faster.
- *Operating system* was used.

Some types of third generation computer from IBM (Mainframe) were:

- IBM System/360
- IBM System/370
- IBM System/4300
- IBM System/3900
- Others were the series of PDP from DEC

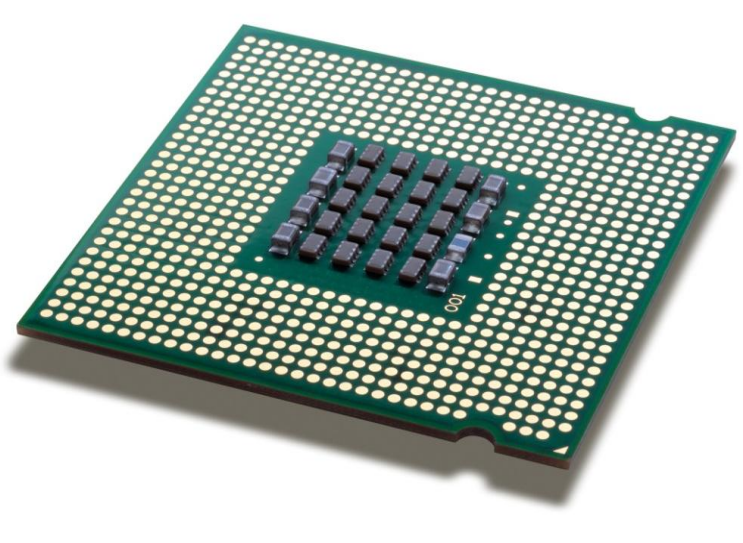


Fourth Generation Computer (VLSI): 1980 to now

In 1980, technology of VLSI (*very large scale integration*) has enabled the use of hundreds and finally millions of transistors in one *chip or micro processor*.

This technology enables the creation of CPU, memory and other components in a single chip which can be produced massively in a cheap price.

At last, it can produce smaller, faster, and cheaper computers, that everyone can afford. This technology starts the era of Personal Computer (PC).



Fifth Generation Computer

The term fifth generation computer is created by a Japanese to describe a “smart” computer which was built in the mid 1990s.

The development involves artificial intelligence, expert system, and natural programming language.

The focus of the fifth generation is connectivity, by connecting a computer to other computers, to form parallel computation.

Six Eras of IT

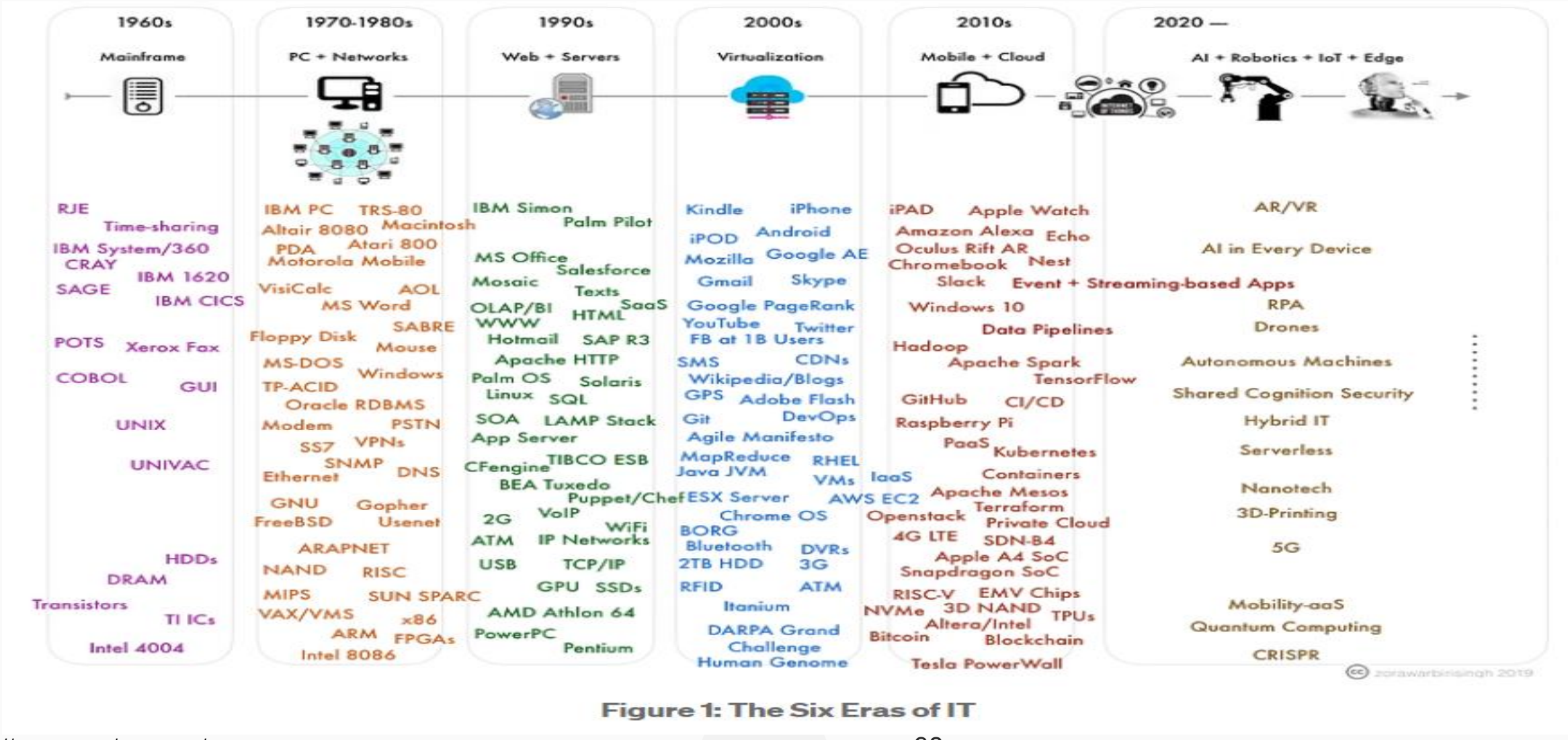


Figure 1: The Six Eras of IT

https://medium.com/@zbirisingh/six-eras-of-it-thoughts-on-technology-and-platform-cycles-921d9b2fae77

Mainframe

- The introduction of the IBM 1401 and 7090 transistorized machines in 1959 marked the beginning of widespread commercial use of **mainframe** computers. In 1965, the mainframe computer truly came into its own with the introduction of the IBM 360 series. Mainframe computers became powerful enough to support thousands of online remote terminals connected to the centralized mainframe using proprietary communication protocols and proprietary data lines.



Personal Computer & Networking

- Proliferation of PCs in the 1980s and early 1990s launched a spate of personal desktop productivity software tools—word processors, spreadsheets, electronic presentation software, and small data management programs—that were very valuable to both home and corporate users. These PCs were stand-alone systems until PC operating system software in the 1990s made it possible to link them into networks.



Networking is the exchange of information and ideas among people with a common profession or other interest, usually in an informal social setting.



Web

What is WWW?
WWW, which stands for World Wide Web, is a subset of the internet. It is also known as W3 or web. WWW refers to all the web pages or online content that is available on the internet in HTML formatted manner. The web pages and documents available on the internet are connected via hyperlinks.

The information available on the web pages can be of various formats consisting of text, image, audio, and video. The WWW or World Wide Web was developed by Tim Berners-Lee at CERN (European library for Nuclear Research) in 1989.



The World Wide Web

World's First Website Goes Online

The first-ever URL was created on December 20, 1990, and brought the world world the first website. It only had plain text and basic basic HTML format weighing weighing in at only 2.7KB.

Netscape's Navigator

In 1994, Netscape Communications developed developed the Navigator browser that incorporated incorporated graphics and and text. The development of development of the web browser marked a change in change in the way people experienced the internet.

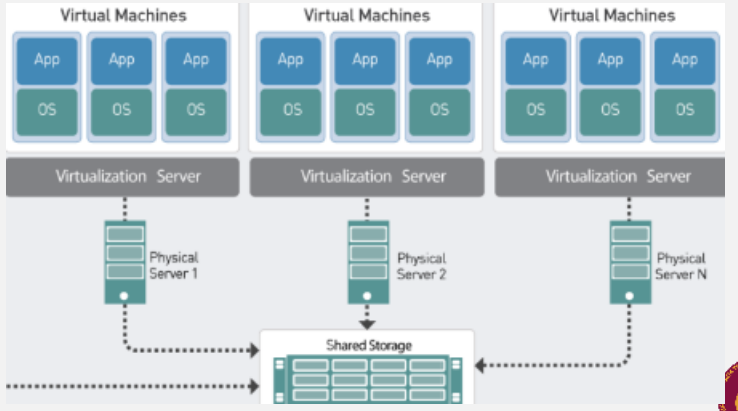
Google Smart Search Algorithm

Launched in 1998, Google's Google's search engine had a had a straightforward interface, making it work a a lot faster than its competitors. It was also the the first search engine that that used PageRank algorithm.

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Virtualization

Virtualization enables the hardware resources of a single computer—processors, memory, storage and more—to be divided into multiple virtual computers, called virtual machines (VMs).



Mobile Computing

Mobile Computing is the technology used for transmitting voice and data through small, portable devices using wireless enabled networks.

Mobile Computing System is a distributed system, which is connected via a wireless network for communication. The clients or the nodes possess mobility and the ability to provide computing at anytime, anywhere.



The Rise of Mobile Technology

1

First Cell Phone Call

In 1973, Motorola researcher Martin Cooper made the first-ever cellular phone call to his rival to his rival research head at Bell Labs, Joel Engel.

2

Introduction of iPhone

On June 29, 2007, Steve Jobs unveiled the first iPhone, altering the way people access the access the internet, take photos, and make phone calls.

3

Mobile Payments

Launched in 2011 as Google Wallet, mobile payments have transformed the payment industry payment industry by allowing people to make payments with a mobile app and avoid carrying avoid carrying cash and cards.

4

Virtual Assistants

Virtual assistants like Siri, Alexa, and Google Assistant emerged in 2011 and quickly gained gained popularity. They have revolutionized the way people interact with their technology technology devices.

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Cloud Computing

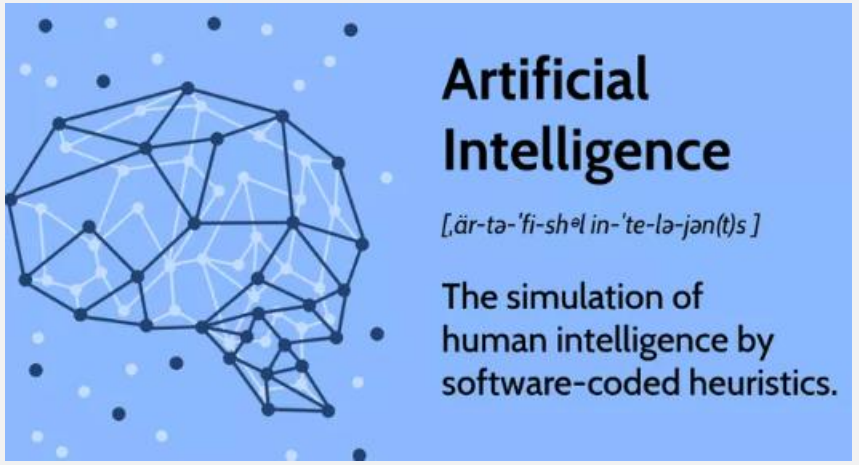
Cloud computing is the **on-demand** delivery of compute power, database, storage, applications, and other IT resources **via the internet** with **pay-as-you-go** pricing.

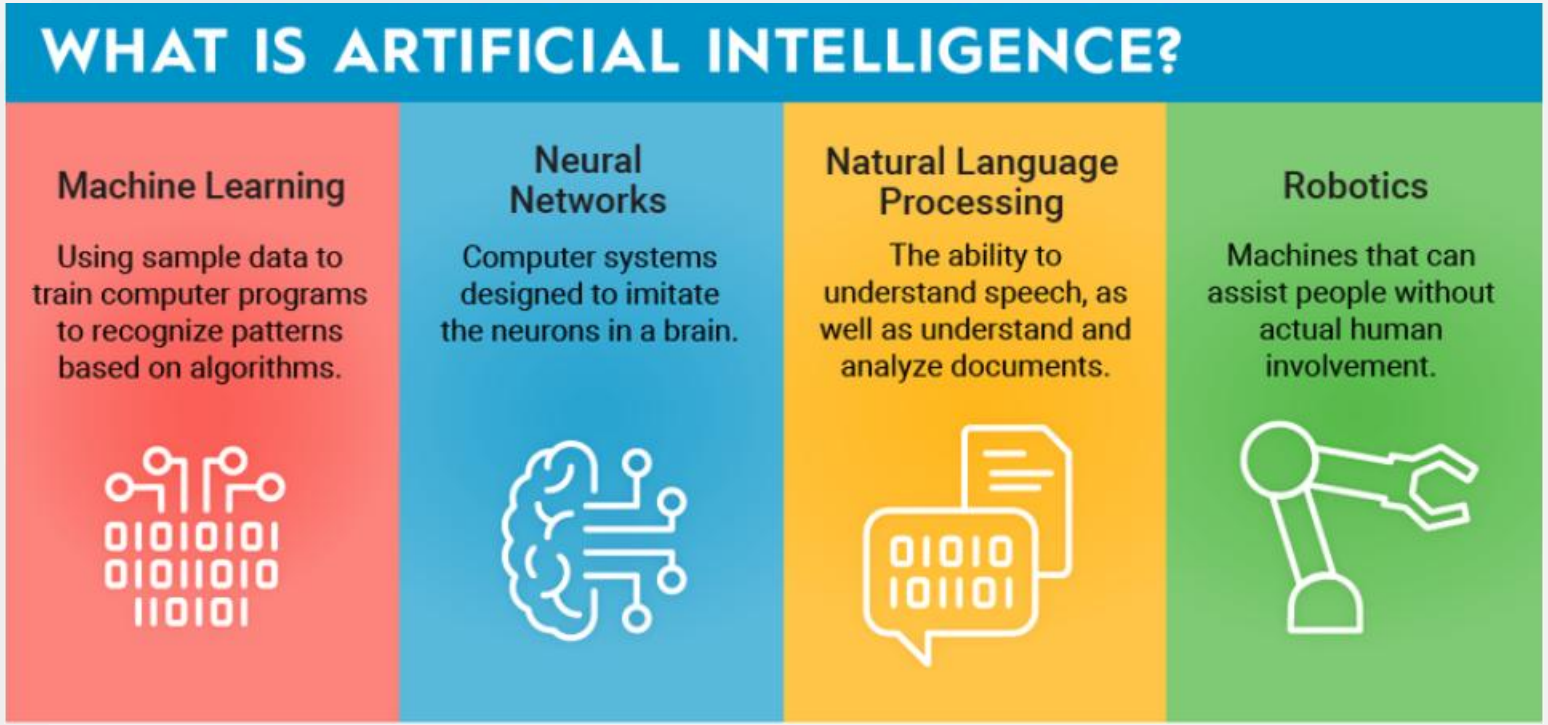


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Artificial Intelligence

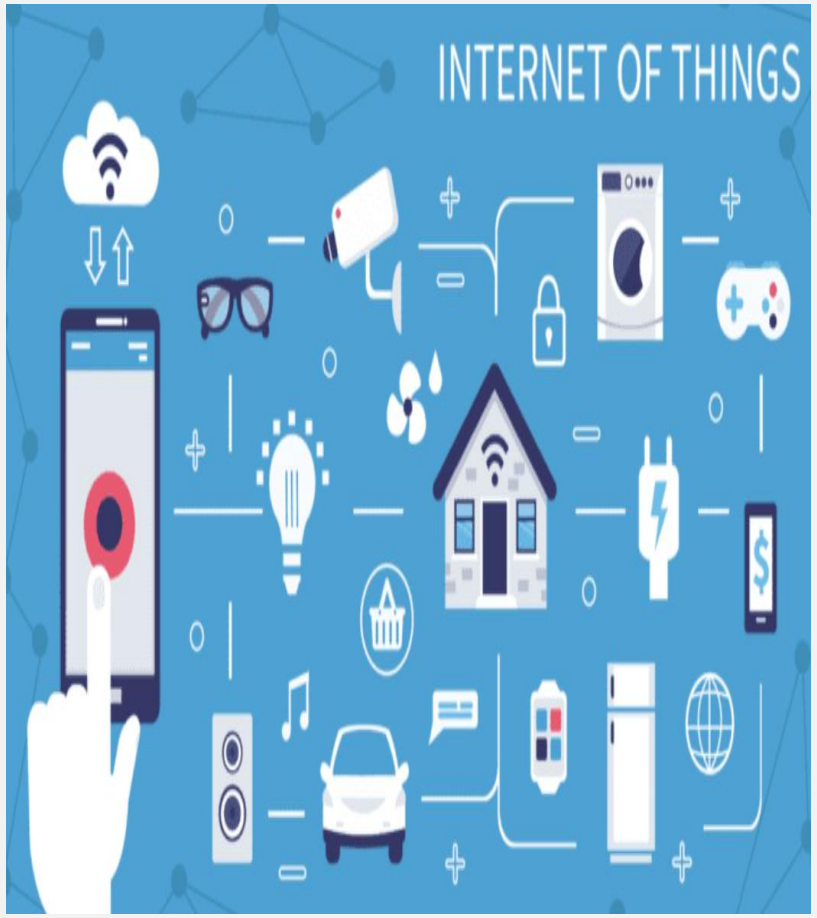
Artificial intelligence is the simulation of human intelligence processes by machines, especially computer systems.





Internet of Things (IoT)

The Internet of Things (IoT) primarily refers to any device, appliance, instrument, or machine that can connect to the internet. They contain sensors, software, and other technologies used to link and exchange data between systems and devices over the Internet.



Internet of Things (IoT)

The Internet of Things (IoT) is a network of interconnected devices that are embedded with sensors, software, and software, and other technologies that enable them to exchange data and interact with other devices and systems. and systems.

1

Smart Homes

IoT enabled devices are transforming transforming homes into intelligent and and connected spaces by allowing devices devices to communicate and exchange exchange data.

2

Smart Cities

The use of IoT in urban planning is enabling cities to become smarter and and more efficient by optimizing traffic traffic flow, reducing energy consumption, consumption, and improving public services.

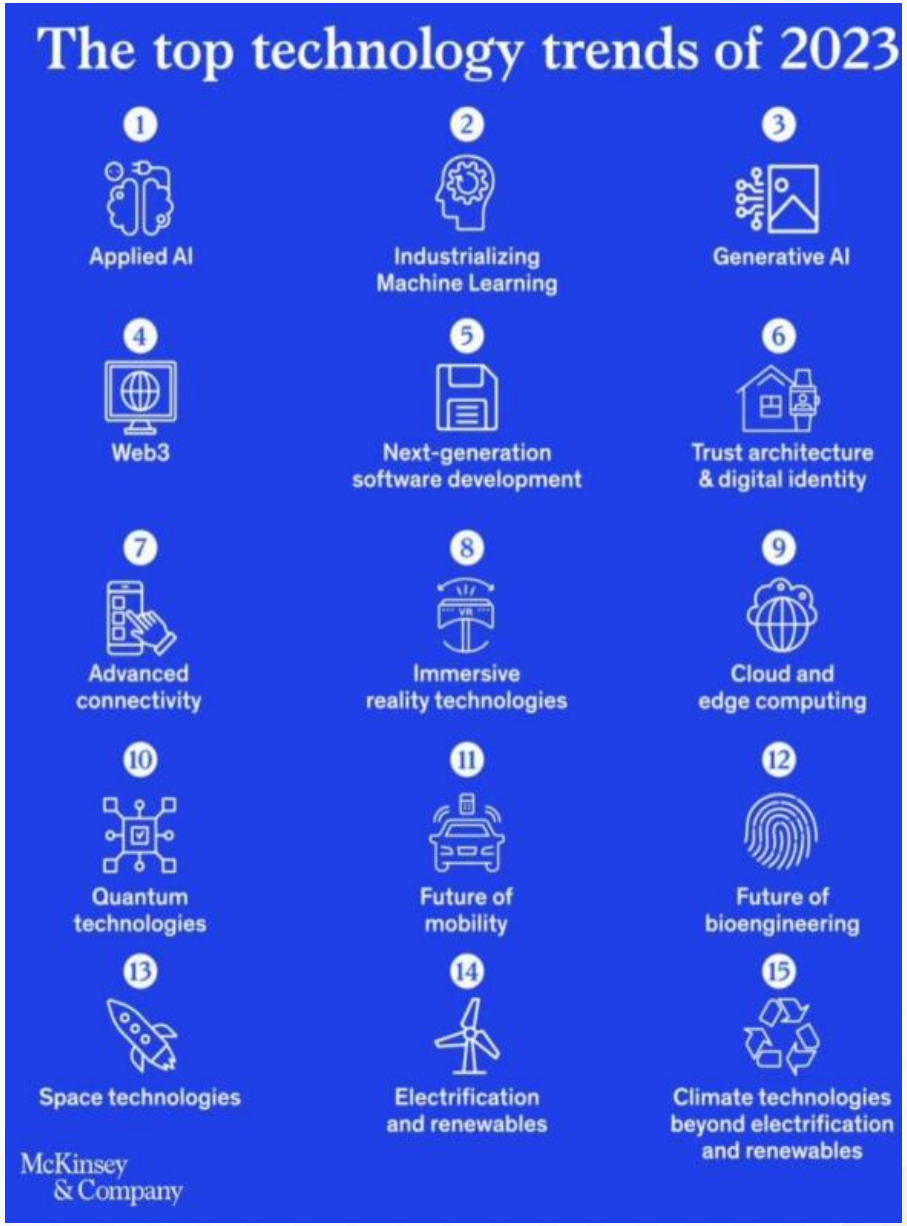
3

Industry 4.0

The use of IoT in manufacturing is creating a new era of automation and and optimization through predictive maintenance, real-time optimization, and optimization, and reduced downtime. downtime.

Edge Computing

A decentralized computing infrastructure that brings compute resources closer to the end-user. This technology reduces latency and increases computation efficiency.


The top technology trends of 2023
1 Applied AI
2 Industrializing Machine Learning
3 Generative AI
4 Web3
5 Next-generation software development
6 Trust architecture & digital identity
7 Advanced connectivity
8 Immersive reality technologies
9 Cloud and edge computing
10 Quantum technologies
11 Future of mobility
12 Future of bioengineering
13 Space technologies
14 Electrification and renewables
15 Climate technologies beyond electrification and renewables
McKinsey & Company


Top Strategic Technology Trends 2023
1 Digital Immune System
2 Applied Observability
3 AI TRiSM
4 Industry Cloud Platforms
5 Platform Engineering
6 Wireless-Value Realization
7 Superapps
8 Adaptive AI
9 Metaverse
10 Sustainable Technology
gartner.com
Source: Gartner
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https://www.founderjar.com/new-technology-trends/

Implications for the Future

As emerging technologies continue to evolve and transform industries, the future holds many possibilities. possibilities.



Architecture and Construction

The use of emerging technologies such as AR, VR, and AI in architecture and construction is revolutionizing how buildings are designed, constructed and maintained.



Transportation

The use of autonomous vehicles, smart traffic management systems, and hyperloop technology is transforming transportation systems and reducing congestion while improving safety and efficiency.

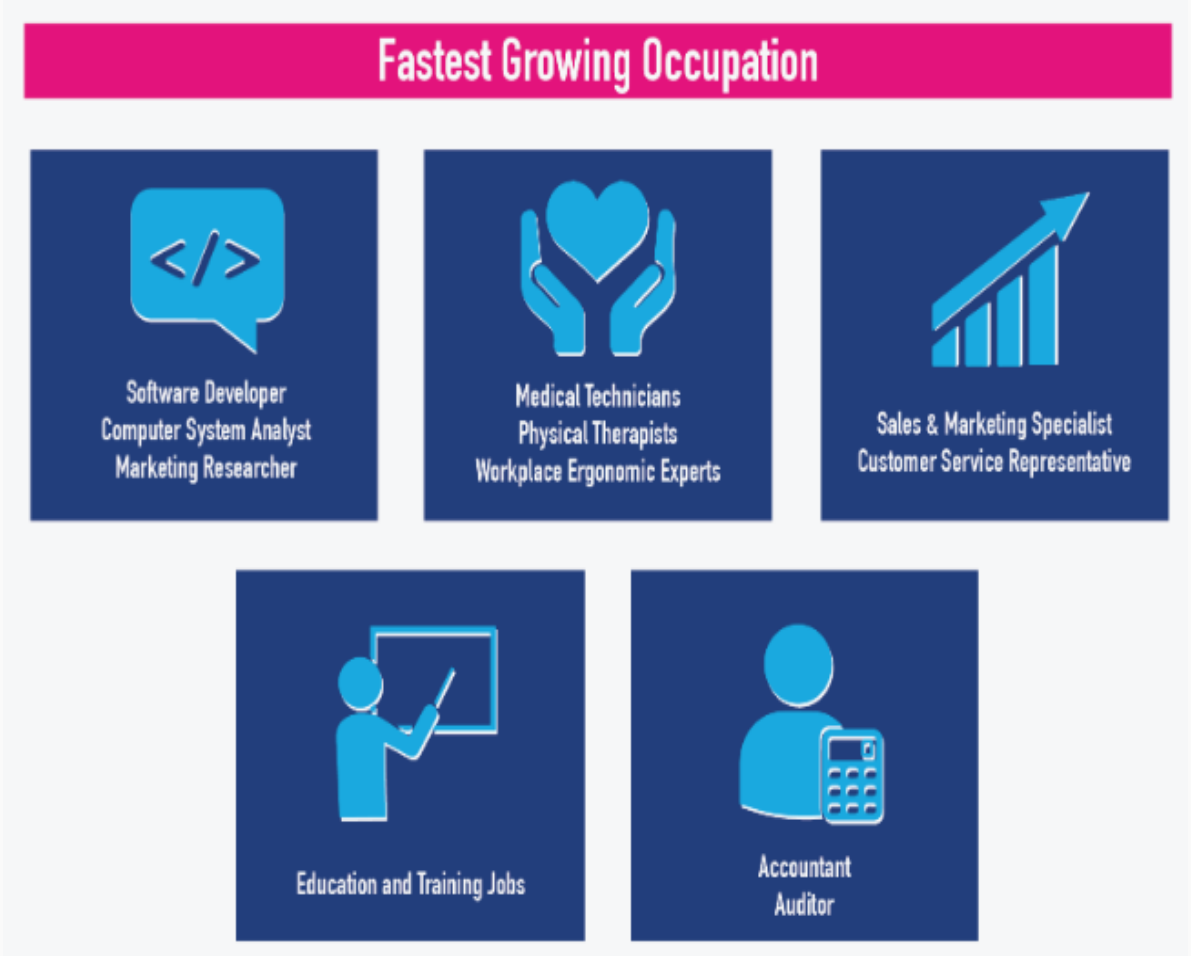


Healthcare

The use of personalized medicine, gene editing, and bioengineering technologies is is accelerating treatments for chronic and genetic diseases and and transforming the healthcare healthcare industry.

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Jobs in 2025







https://www.jobstreet.com.sg/career-advice/article/jobs-2025-expect-survive

THANK YOU

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In the Name of God for Mankind

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